

RETRO

Functional Specification

Version 1.0

Document Authors:
Chun Lung Lim
Cindy Burklow
Jody Larsen

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

Scope

This document outlines the requirements for RETRO-Beta software.

Responsibility

Document Authors

Chun Lung Lim, Cindy Burklow, Jody Larsen

Document Maintainers

SVVRL Team

Contributors

Chun Lung Lim, Cindy Burklow, Jody Larsen, SVVRL Team

Approvals

SVVRL Team

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

Revision History

Date	Version	Description	Author
09/01/2006	0.1	First draft – Design layout and set requirements.	Chun Lung Lim Cindy Burklow Jody Larsen
09/02/2006	0.2	Second draft – Review RETRO documents.	Chun Lung Lim Cindy Burklow Jody Larsen
09/03/2006	0.3	Third draft – Review RETRO applications	Chun Lung Lim Cindy Burklow Jody Larsen
09/04/2006	0.4	Fourth draft – Filling all the information for this requirement document.	Chun Lung Lim Cindy Burklow Jody Larsen
09/05/2006	0.5	Fifth – Continue filling all the information for this requirement document.	Chun Lung Lim Cindy Burklow Jody Larsen
09/06/2006	0.6	Sixth – Working on test cases.	Chun Lung Lim Cindy Burklow
09/07/2006	0.7	Seventh – Working on test cases.	Jody Larsen
09/10/2006	1.0A	Alpha final review	Chun Lung Lim Cindy Burklow Jody Larsen
09/11/2006	1.0B	Quoted thesis text blocks	Jody Larsen
09/22/2006	1.0C	Reformatted/renamed document as a baseline for SVVRL RETRO development	Jane Hayes

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

Table of Contents

Document Owner	2
Document Maintainer	2
Contributors	2
Approvals	2
 1. Introduction	 7
1.1 Purpose	7
1.2 Scope	7
1.3 Definitions, Acronyms, and Abbreviations	7
1.4 References	7
1.5 Overview	8
 2. Overall Description	 8
2.1 Product Perspective	8
2.2 Background and Related Work	9
2.2.1 Tracing Process	9
2.2.2 Automated Tracing	10
2.2.3 Information Retrieval Methods	10
2.2.4 Feedback	11
2.2.5 Thesaurus	11
2.3 Product Functions	12
2.4 User Characteristics	12
2.5 Operating System, Devices and Other Constraints	12
2.6 Assumptions and Dependencies	12
 3. System Features	 14
3.1 START NEW PROJECT	14
3.1.1 Description and Priority	14
3.1.2 Functional Requirements	14
3.2 Load Project	14
3.2.1 Description and Priority	14
3.2.2 Functional Requirements	15
3.3 LOAD RTM	15
3.3.1 Description and Priority	15
3.3.2 Functional Requirements	15
3.4 SAVE	16
3.4.1 Description and Priority	16
3.4.2 Functional Requirements	16
3.5 CLOSE CURRENT PROJECT	16
3.5.1 Description and Priority	16
3.5.2 Functional Requirements	16
3.6 EXIT	16
3.6.1 Description and Priority	16
3.6.2 Functional Requirements	16
3.7 SHOW XML	17

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

3.7.1	Description and Priority	17
3.7.2	Functional Requirements	17
3.8	GENERATE REPORT	17
3.8.1	Description and Priority	17
3.8.2	Functional Requirements	17
3.9	FILTER LOW LEVEL REQUIREMENTS	17
3.9.1	Description and Priority	17
3.9.2	Functional Requirements	17
3.10	DISPLAY LOW LEVEL ELEMENTS	18
3.10.1	Description and Priority	18
3.10.2	Functional Requirements	18
3.11	FEEDBACK	18
3.11.1	Description and Priority	18
3.11.2	Functional Requirements	18
3.12	MULTIPLE PROJECTS	18
3.12.1	Description and Priority	18
3.12.2	Functional Requirements	18
3.13	IMPORT	19
3.13.1	Description and Priority	19
3.13.2	Functional Requirements	19
3.14	GUI INTERFACE	19
3.14.1	Description and Priority	19
3.14.2	Functional Requirements	19
4.	BROWSE MODE	19
4.1	KEYWORD SEARCH	19
4.1.1	Description and Priority	19
4.1.2	Functional Requirements	19
4.2	MANUAL LINKING/TRACING	19
4.2.1	Description and Priority	19
4.2.2	Functional Requirements	19
5.	TRACE MODE	19
5.1	TRACING	19
5.1.1	Description and Priority	19
5.1.2	Functional Requirements	20
5.2	TRACING RESULTS	20
5.2.1	Description and Priority	20
5.2.2	Functional Requirements	20
5.3	SAVE TRACE	20
5.3.1	Description and Priority	20
5.3.2	Functional Requirements	20
5.4	COMPLETING TRACE	20
5.4.1	Description and Priority	20
5.4.2	Functional Requirements	20
6.	ASSESS MODE	20
6.1	RUN ACCURACY & COMPLETENESS	20

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

6.1.1 Description and Priority	20
6.2 SHOW RECOMMENDATIONS LIST	21
6.2.1 Description and Priority	21
6.2.2 Functional Requirements	21
6.3 RUN SATISFACTION ANALYSIS	21
6.3.1 Description and Priority	21
6.3.2 Functional Requirements	21
6.4 REMOVE ALL NO LINKS	21
6.4.1 Description and Priority	21
6.4.2 Functional Requirements	21
6.5 DELETE RECOMMENDED LINKS	22
6.5.1 Description and Priority	22
6.5.2 Functional Requirements	22
6.6 COLOR-CODED SATISFACTION STATUS	22
6.6.1 Description and Priority	22
6.6.2 Functional Requirements	22
7. Non-Functional Requirements	22
7.1 Usability	22
7.2 Reliability	22
7.3 Performance	22
7.4 Supportability	22
7.5 Design Constraints	22
7.6 User Documentation Requirements	22
7.7 Interfaces	23
7.7.1 User Interfaces	23
7.7.2 Hardware Interfaces	23
7.7.3 Software Interfaces	23
7.7.4 Communications Interfaces	23
7.8 Licensing Requirements	23
7.9 Legal, Copyright, and Other Notices	23
7.10 Applicable Standards	23
8. Supporting Information	23
8.1 Requirements/RETRO Traceability Matrix	23
APPENDIX 1 – Test Cases For Assess Mode	24

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

Functional Specification

1. Introduction

1.1 Purpose

This functional specification describes the software functional and nonfunctional requirements for release V.Beta of the RETRO (Requirements Tracing On-Target) application. This document is intended to be used by the members of the project team that will implement and verify the correct functioning of the module. Unless otherwise noted, all requirements specified here are high priorities and committed for release V.Beta.

1.2 Scope

The RETRO tool is a software application intended to assist a user in analyzing the relationship between two requirement documents, one providing a high level of requirements, the other a low level. It provides the ability to generate a requirement traceability matrix (RTM) as a result of this analysis or to validate an existing RTM. By providing a formal tool for RTM management, both the accuracy and the details contained in the RTM can be improved.

1.3 Definitions, Acronyms, and Abbreviations

RETRO	RE quirements TR acing On target
RTM	Requirements Traceability Matrix
IR	Information Retrieval

1.4 References

The following documents were used in the creation of this document or are referenced by this document:

1. S. Vadlamudi, *RETRO (REquirement TRacing On target) Masters Project Report*, University of Kentucky, CS Masters Program, 2005.
2. L.A. Maciaszek, *Requirements Analysis and System Design, Second Edition*, pp. 47-75, Boston, MA: Pearson Addison-Wesley, 2005.
3. J. H. Hayes, *RTM, RETRO, and Requirements*, University of Kentucky CS616 Lecture, August 31, 2006.
4. J. H. Hayes, *RETRO Documentation*, RETRO-usecases.doc, 2005.
5. RETRO Users Manual. http://sweb.uky.edu/~svadl2/Users_Manual.html

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

1.5 Overview

The rest of the functional specification provides a general description of the project. It includes a detailed description of the functions and features. It also describes the non-functional requirements such as design constraints, assumptions, and dependencies.

2. Overall Description

2.1 Product Perspective

Introduction

"Tracing of requirements has been a necessary and demanding task in the software development life cycle. Analysts spend most of their time in building traceability matrices to demonstrate their designs satisfying the requirements. Trace can be built in two ways, one is starting from the customers statements of work and the other building the traceability information as the process unfolds. The latter practice is known as after-the-fact tracing which is the most widely used method for building knowledge base or Independent Verification and Validation.

The purpose of Independent Verification and Validation (IV & V) is to make sure that the right process has been used to build the right system (verification) and the right system was built (validation). During each phase of the software life cycle the process must be verified and all the requirements that have been implemented at the end of the life cycle must be validated. So, the requirements traceability matrix (RTM) can be viewed as the backbone of IV & V. The requirements traceability matrix maps high level artifacts such as requirements with the low level design elements. The RTM play a vital role in predicting software's quality before the system is built. The development process requires a lot of analyst time because the high level and low level documents must be parsed and analysts must manually discover the links between the high level artifacts and low level artifacts. There are many semi-automated tools for building RTM but most of them require analyst involvement. The current approaches used for tracing requires analysts performing interactive searches for potential linking requirements. The analyst needs to manually find keywords and assign them prior to tracing. With these approaches the correct links evaluation is based on analyst decision and they may not be accurate because of the possible human error. The analyst needs to scan trough the built trace and need to redo the above process if the answer set is not accurate which leads to the tracing process becoming more complex. So, some methods need to be developed to generate the candidate links by automating the process of keyword assigning and parsing the documents. An effective and efficient tool to overcome the above problem can be designed using the Information Retrieval (IR) techniques.

RETRO (Requirements Tracing On-target) is the tool developed with the information retrieval techniques. The first version of the tool was designed and proved very efficient in building the requirements tracing matrix. RETRO had very good back-end but the front-end demanded a wider range of facilities for tracing. So, this necessitated for the development of a new GUI. The features which the new GUI demanded were two important modes of operations. The first one was to build a new RTM with the help of the back end methods written in C++. The other mode was to assess an existing RTM. Both of these modes provide a tab to support the manual tracing to add any missing links with the help of browse tab. Filtering functionality which was part of the back-end architecture in the old version is to be made available to the analyst for displaying of candidate links according to his requirement." [1]

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

RTM Assessment Process

“RTM plays an important role not only in the process of tracing the requirements but also when the product undergoes any changes. When a change is made to a product the RTM helps in knowing the impact of the change on the process. It also helps in prioritizing the requirements by their importance and helps in identifying the requirements that are to be tested effectively. In addition the requirements change frequently as the process develops, thus requires a change in the RTM. Making a change to the requirement is very frequently done during the process of development. So, an automated method for updating the RTM is necessary.” [1]

“Most of the times the analyst is provided with an RTM by the developer to evaluate the quality and completeness. So, the properties of the given RTM accuracy and completeness along with the satisfaction of the underline document must be determined to assess a given RTM. Many of the existing tools do not provide the option of assessing an RTM. Very less research was done in the area of assessing an RTM and most of them found using information retrieval and data mining strategies can be helpful. So, by using the already existing IR methods that are developed for RETRO the accuracy and completeness of any given RTM is evaluated and potential links for that RTM which are not part of it are also suggested. RETRO also provided the satisfaction analysis with the help of natural language (NLP) techniques which shows how far a requirement is satisfied by the list of its generated candidate links.” [1]

2.2 Background and Related Work

“Independent verification and validation (IV & V) is a process done to confirm the product developed or being developed is correct, complete and consistent with the specifications. The independent verification and validation is a process with a set of performance tests to analyze the process of software development and operation of the developed software. The most popularly used approach by IV & V is a requirements driven methodology. In this methodology the functionality of the product is expressed as requirements. Initially all the requirements are identified. Then all the requirements are matched with the corresponding designing elements and a tracing matrix is built. This trace matrix is used for the further analyses of the product. The whole process is known as tracing process which in detail is shown below.” [1]

2.2.1 Tracing Process

“The process of tracing is done in three steps. First the analyst gets the document which records all the user specifications and breaks them into the requirements of the system that is being built. After identifying the requirements the second step is to analyze them. To analyze the requirements, the analyst needs to identify keywords in the high level and low level documents and a matching strategy should be applied to generate the matching or candidate links which is nothing but a list of low level requirements that satisfy the corresponding high levels. The candidate links generation is done in wide range of different ways and this takes a lot of time because building an initial RTM needs to be done very carefully and effectively as it affects the whole of remaining process. Finally after the candidate links are generated they are evaluated by the analyst in the next step. The analyst scans the generated candidate links and finds the true links and false links. False links are the links that are not valid for the given high level requirement. The analyst thus provides the feedback on each of the candidate link by stating it to be a true or false link. Then the feedback from the analyst is taken and the regeneration of candidate links is done using the same matching algorithm of step 2. The result is again evaluated by the analyst and this process is repeated until a valid trace is built.” [1]

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

2.2.2 Automated Tracing

“Tracing of requirements needs a lot of routine actions to be done and demands a lot user interaction. The time spent on this tracing process can be reduced by applying automated tools. But the process of tracing cannot be automated completely. The process of breaking requirements from user specifications and evaluation of the generated candidate links needs to be done manually by analyst. But the process of generating candidate list can be completely automated to decrease the analyst's total time spent in the tracing. A lot of tools are developed to support the automated tracing activity. But the analyst needs to provide the keywords by looking the two level documents or needs to do the matching process manually to generate the candidate links or generates a lot of false links. So, all the existing tools still needs the analysts interaction. A requirement tracing tool with the automated process is defined as one which takes the requirements as inputs and outputs the traceability matrix which can be evaluated by the analyst. RETRO is the requirement tracing tool that's built for the above automated process. RETRO uses the Information Retrieval (IR) and text mining methods to build the lists of candidate links. It also provides option of building the candidate links with a thesaurus and stop word file. The backbone of the tool is the information retrieval methods. The aim of these IR methods in RETRO is to find the set of documents that match the given query.” [1]

2.2.3 Information Retrieval Methods

“The working of any Information Retrieval method is done in two stages. First, the documents are analyzed and indexed. Then the system analyzes the incoming queries and uses a matching algorithm to determine whether the documents are similar to the query representations. When the IR is applied to the traceability process high level document becomes the query and the low level documents are the documents which are returned as a result after the process is finished.

Vector space retrieval

Vector space retrieval methods represent each document and each query as a vector of keyword weights. These keywords are words present in the given documents. Each document is passed through a stop-word file to avoid the most common words in finding the keywords. When a query is given the most common words like “The” etc they are written in a stop word file which eliminates them in the building process. But some projects may have some specific common words like “error” in software related projects. These can be added to the stop word file to make an effective building.

The vocabulary of the document collection, $D = \{k_1, \dots, k_N\}$, is formed as the union of all terms found in all documents. Each document, d_i , is then represented as a vector, $di=(w_{i1}, \dots, w_{iK})$, of term/keyword weights. Different weight schemes are proposed for building the vector of keywords in IR techniques. Most widely used of them is Tf-idf the standard term-weighting computation method and uses the

formula $w_{ij} = tf_{ij} \cdot \log\left(\frac{n}{df_j}\right)$, where tf_{ij} is the term frequency (Tf) of keyword k_j in document d_i , is the

normalized frequency of occurrence of k_j in d_i , while $\log\left(\frac{n}{df_j}\right)$, is called the inverse document

frequency(Idf) of term k_j . These terms measure how rarely the term is found in the entire document and builds the document and query vectors. These vectors are used as base to compute the similarity between them as the cosine angle of the angle between them.

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

$$\text{sim}(d, q) = \cos(d, q) = \frac{\sum_{j=1}^N d_j \cdot q_j}{\sum_{j=1}^N d_j^2 \cdot \sum_{j=1}^N q_j^2}$$

The cosine value is used as the weight value that is the relativity of the given document with the current query.

The above process in terms of high level documents and low level documents can be explained as follows. Every high level document is compared with all low level documents using the term frequency and inverse document frequency to give the set of valid low level links. All valid low level links are displayed along with the weights calculated using the cosine angle. These are analyzed by the analyst and his feedback is considered to recalculate the valid links.” [1]

2.2.4 Feedback

“The process of feedback starts after the candidate links are generated. They are evaluated by the analyst by providing positive feedback and negative feedback. This analysis of user is the input for regenerating the candidate links. The way the feedback works for generating the new candidate links is, if there is positive feedback for any link all links related to that are given a increase in their weights that is the keywords found in that document increase the value in query vector and if a link is given a negative feedback all related links are given a decrease in weights then the keywords found in the document decrease the value in query vector. Let Dq be a list of document vectors retrieved. The user feedback identifies two subsets in Dq : Dr of size R documents relevant to q and $Dirr$ of size S of irrelevant documents. Dr and $Dirr$ are disjoint, but do not necessarily cover the entire set Dq . We use Standard Rochio [Baeza-Yates] feedback processing method:

$$q_{new} = \alpha q + \left(\frac{\beta}{r} \sum_{d_j \in Dr} d_j \right) - \left(\frac{\gamma}{s} \sum_{d_k \in Dirr} d_k \right)$$

Here, query q is adjusted by adding to its vector a vector consisting of the document vectors identified as relevant, and subtracting from it the sum of all document vectors identified as false positives. The constants α , β , γ in the formulas above can be adjusted in order to emphasize positive or negative feedback as well as the importance of the original query vector. Once the query vectors have been recomputed, the selected IR method is applied to the modified query vectors. This loop can be repeated until the user is satisfied with the results.” [1]

2.2.5 Thesaurus

“To make the trace process more effective we use the thesaurus and stop word file. In any project the documents that are traced are written by a lot of different people. Different people may use different names for the same term. For example an artifact may have a word “error” whereas the other might use the same as “failure”. IR techniques that are used in tracing do not relate the above both words and consider them to be different even tough they mean the same. To address this problem RETRO provides an option of providing Thesaurus which takes a set of triples (v, a, w) where v and w are the words that are relevant and a shows how closely they are related. The analyst can provide these thesaurus files by enabling the option and providing the location to built file. When the IR method is ran

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

with the enabled thesaurus option it goes through an extra step to adapt to the situation. First all the thesaurus terms that are not keywords are made separate keywords and the cosine is calculated by adding the product of their weights and similarity measure along with the standard formula.

All the above options are the advanced option used to make the process of building the candidate links more effective and simpler. These are the options that can be selected by the analyst to build a trace in more effective way. All these inputs are sent to the back end process which computes the result set.”
[1]

2.3 Product Functions

The major functionality covered includes document parsing (of both the requirements document and the RTM file), candidate link generation, candidate link evaluation, and analysis of RTM accuracy.

2.4 User Characteristics

RETRO has only one user class, a software requirements analyst.

2.5 Operating System, Devices and Other Constraints

- OE-1: RETRO shall run on MS Windows 2000 and MS Windows XP.
- OE-2: RETRO shall run using Java version 1.4 and later.

2.6 Assumptions and Dependencies

Interaction of RETRO front end with the IR methods

“RETRO front end allows analyst to use a variety of IR methods. They are Vector Space Retrieval, Keyword Extraction and Latent Semantic Indexing. LSI is only implemented for Linux version so it's not working in the current designed version.

The interaction of RETRO front end with the IR methods can be explained with the help of architecture while the interaction of the process is explained with the help of Vector Space Retrieval method.

Initially if the tracing process is called for the first time the build method is called which constructs the vectors. Once the vectors are built they are sent to the IR tool box where the “Vector Space Retrieval” technique is applied and the all the links that are related to high levels are built and their weights are calculated using the tf-idf formulas. After that they are ranked by their weights and a resultant XML file is sent to the front end. The front end then parses the file and displays the result on the GUI. The analyst submits the feedback on the evaluated candidate links. When a feedback is submitted the feedback processor handles the request where the weights are recalculated depending on the type of feedback. Then the “XML” file is updated with the modified query representations and is sent to the IR toolbox. There the “Vector Space Retrieval” technique is again applied to compute the new candidate links and send the result to the front end where they are parsed and again displayed for the user analysis. This process can be repeated in a loop until the tracing process is complete.” [1]

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

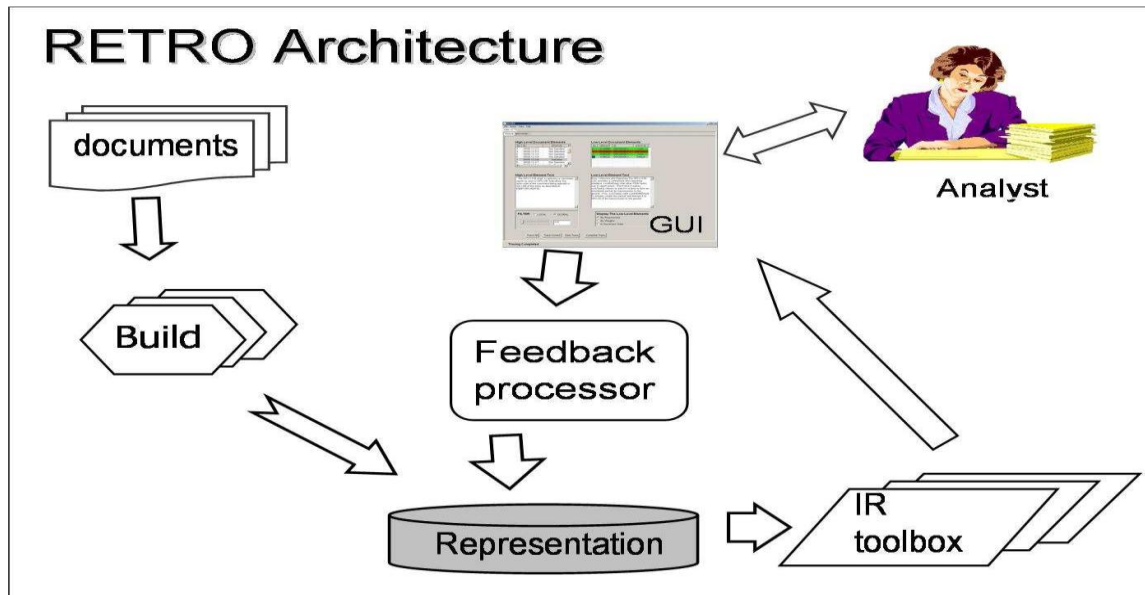


Figure 1: RETRO Architecture [1]

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

3. System Features

3.1 START NEW PROJECT

3.1.1 Description and Priority

The **START NEW PROJECT** function shall allow the user to create a new project.

3.1.2 Functional Requirements

3.1.2.1 The user shall be prompted to provide the new project name.

3.1.2.2 The user shall be prompted to provide the low level document path.

3.1.2.3 The user shall be prompted to provide the high level document.

3.1.2.4 The user shall optionally be prompted to select an IR Method from the following list:

- Vector-Space Retrieval
- Keyword Extraction
- Keyword Extraction-IDF

3.1.2.5 The user shall optionally be prompted to select a Vocabulary Base of either "Both Documents" or "Low Level Document Only". The default value is "Both Documents".

3.1.2.6 The user shall optionally be prompted to provide Weight Option from the following list:

- TF-IDF (Default)
- Okapi
- LTU

3.1.2.7 If the user has selected an IR Method of Keyword Extraction or Keyword Extraction-IDF, the user must also select a Weight Threshold.

3.1.2.8 The user shall optionally be prompted to provide a Thesaurus Filename.

3.1.2.9 The user shall optionally be prompted to provide the Stop Word file.

3.1.2.10 The user shall optionally be prompted to provide a Feedback Method. "Standard Rochio" is currently the only option.

3.1.2.11 The user shall be able to cancel the process and return to the previous state at any time before the project is created.

3.1.2.12 The user shall be able to create the project using the above input. The project shall be placed in "Trace" mode.

3.2 Load Project

3.2.1 Description and Priority

Since the majority of projects are not completed in a single setting, then it is critical to provide the user the option to load an existing project to continue working on the project. Load Project feature will provide the user the capacity to open an existing project in the last saved state with all the associated mode options.

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

[1]

3.2.2 Functional Requirements

3.2.2.1 The user shall be prompted to select a project name from a list of saved projects. Project names are obtained by scanning the saved projects directory.

3.2.2.2 The user shall be prompted to select a feedback method. The only option is "Standard Rochio".

3.2.2.3 The user shall be able to cancel the **LOAD PROJECT** action and return to the previous activity.

3.2.2.4 The user shall be able to load the project once the project name and feedback method is selected.

3.3 LOAD RTM

3.3.1 Description and Priority

Similar in functionality to the **START NEW PROJECT** function, the **LOAD RTM** function collects project parameters from the user and initializes the project. A project created via the **LOAD RTM** function may be in either "trace" or "assess" mode. Assess mode is usually used when the user is provided a RTM and needs to evaluate its accuracy, completeness, and satisfaction level.

3.3.2 Functional Requirements

3.3.2.1 The user shall be prompted to provide the new project name.

3.3.2.2 The user shall be prompted to provide the low level document path.

3.3.2.3 The user shall be prompted to provide the high level document path.

3.3.2.4 The user shall be prompted to provide the RTM file path.

3.3.2.5 The user shall optionally be prompted to select an IR Method from the following list:

- Vector-Space Retrieval
- Keyword Extraction
- Keyword Extraction-IDF

3.3.2.6 The user shall optionally be prompted to select a Vocabulary Base of either "Both Documents" or "Low Level Document Only". The default value is "Both Documents".

3.3.2.7 The user shall optionally be prompted to provide "Weight Option" from the following list:

- TF-IDF (Default)
- Okapi
- LTU

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

3.3.2.8 If the user has selected an IR Method of “Keyword Extraction” or “Keyword Extraction-IDF”, the user must also select a “Weight Threshold”.

3.3.2.9 The user shall optionally be prompted to provide a “Thesaurus” filename.

3.3.2.10 The user shall optionally be prompted to provide the “Stop Word” filename.

3.3.2.11 The user shall optionally be prompted to provide a “Feedback Method”. “Standard Rochio” is currently the only option.

3.3.2.12 The user shall optionally be prompted to provide the “Project Mode” of either “Trace” or “Assess”. “Trace” is the default mode.

3.3.2.13 The user shall be able to cancel the process and return to the previous state at any time before the project is created.

3.3.2.14 The user shall be able to create the project using the above input. The project shall be placed in specified “Project Mode”.

3.4 SAVE

3.4.1 Description and Priority

The user shall be able to activate the **SAVE** function to save the current data and status of the project.

3.4.2 Functional Requirements

3.4.2.1 When activated RETRO will save all current data and project.

3.5 CLOSE CURRENT PROJECT

3.5.1 Description and Priority

The **CLOSE CURRENT PROJECT** Function will allow the user to close a given project without exiting the RETRO application.

3.5.2 Functional Requirements

3.5.2.1 The user shall be able close the current project.

3.6 EXIT

3.6.1 Description and Priority

It is necessary to provide a mechanism to exit the RETRO application when the user is finished working with the application.

3.6.2 Functional Requirements

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

3.6.2.1 The user shall be able to exit the application.

3.7 SHOW XML

3.7.1 Description and Priority

The user may want to view the current project's XML file format. The **SHOW XML** function will provide the user a display of the current XML file.

3.7.2 Functional Requirements

3.7.2.1 The user shall be able to view the XML version of the RTM.

3.8 GENERATE REPORT

3.8.1 Description and Priority

After a project has completed either a trace or an assess process, then the user will need the capability to review the final results of the process in an easy to read format other than the XML file format that is automatically generated throughout the process. The **GENERATE REPORT** function will provide the user a list of reports that the user may view. The user will have the option to save the generated report or modify the design of the report. [1]

3.8.2 Functional Requirements

3.8.2.1 The user shall be able to generate and save the "RTM report"

3.8.2.2 The user shall be able to generate and save the "Parent with no children report"

3.8.2.3 The user shall be able to generate and save the "Parents with no children with text report"

3.8.2.4 The user shall be able to generate and save the "Children with no parents report"

3.8.2.5 The user shall be able to generate and save the "Children with no parents with text report"

3.8.2.6 The user shall be able to generate and save the "Not satisfied requirements report"

3.8.2.7 The user shall be able to generate and save the "Not satisfied requirements with text report"

3.9 FILTER LOW LEVEL REQUIREMENTS

3.9.1 Description and Priority

"The user shall be able to limit the display of low-level requirements or candidate requirement links." [1]

3.9.2 Functional Requirements

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

3.9.2.1 The user shall be able to designate whether the filter process applies to the selected high level document element (LOCAL) or to all high level document elements (GLOBAL).

3.9.2.2 The user shall be able to filter the list of low-level requirements by setting a cutoff point for the weight between 0 and 1.0.

3.9.2.3 The user shall be able to designate that the top N elements to be displayed.

3.9.2.4 The user shall be able activate a **KILL HIDDEN** Links function to remove all elements not currently displayed.

3.10 DISPLAY LOW LEVEL ELEMENTS

3.10.1 Description and Priority

The user will need the ability to view the text of low-level requirements. The **DISPLAY** function will provide three different sort options to view the text of low-level requirements. The user can sort the low-level requirements based upon the weight of linking relationships, original document order, or by requirement id. [1]

3.10.2 Functional Requirements

3.10.2.1 The user shall be able to sort the low level documents by "Requirement ID"

3.10.2.2 The user shall be able to sort the low level documents by "Weights"

3.10.2.3 The user shall be able to sort the low level documents by "Document Order"

3.11 FEEDBACK

3.11.1 Description and Priority

Since RETRO's tracing and assessing modes are based upon algorithms that will allow processes to be improved as the system receives **FEEDBACK** from the user.

3.11.2 Functional Requirements

3.11.2.1 The user shall select a low level requirement and identify the link status from the following list:

- Link
- No Link
- Default

3.12 MULTIPLE PROJECTS

3.12.1 Description and Priority

The user shall be able to load multiple projects at the same time.

3.12.2 Functional Requirements

3.12.2.1 The RETRO System shall allow multiple projects to be open at the same time

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

3.13 IMPORT

3.13.1 Description and Priority

The user shall be able to import projects.

3.13.2 Functional Requirements

3.13.2.1 The RETRO System shall provide a mechanism to convert existing input text files into RETRO file format. (Not implemented.)

3.14 GUI INTERFACE

3.14.1 Description and Priority

The user shall use a **GUI INTERFACE** to work with RETRO.

3.14.2 Functional Requirements

3.14.2.1 The RETRO System shall provide a GUI mechanism for the user to interact with the RETRO System.

4. BROWSE MODE

4.1 KEYWORD SEARCH

4.1.1 Description and Priority

The user is provided the capability via **BROWSE** mode to manually link low-level requirements to high-level requirement. In order to help the user to quickly locate the desired low-level requirement or high-level requirement, a **KEYWORD SEARCH** function is offered to the user. [1]

4.1.2 Functional Requirements

4.1.2.1 The user shall be able to enter a keyword into a search text box to search within low-level requirements and the system shall return matches.

4.1.2.2 The user shall be able to enter a keyword into a search text box to search within high-level requirements and the system shall return matches.

4.2 MANUAL LINKING/TRACING

4.2.1 Description and Priority

There will be times when the user will not agree with RETRO's list of candidate links. Therefore, it is important to provide a mechanism for the user to manually change, create, or delete links between low and high level requirements. **BROWSE** mode is the mechanism within RETRO that enables the user to perform **MANUAL LINKING/TRACING**. [1]

4.2.2 Functional Requirements

4.2.2.1 The user shall be able to create and remove low-level requirements with high-level requirements.

5. TRACE MODE

5.1 TRACING

5.1.1 Description and Priority

One of the purposes of RETRO is to aid the user in linking low-level requirements with high-level

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

requirements. **TRACING** functionality is the component of RETRO that automatically generates a list of candidate links of high to low-level requirements.

5.1.2 Functional Requirements

5.1.2.1 RETRO shall provide a mechanism for the user to generate a trace on all requirements.

5.1.2.2 RETRO shall provide a mechanism for the user to generate a trace on a given requirement(s).

5.2 TRACING RESULTS

5.2.1 Description and Priority

After the **TRACING** function is performed, then the results of the **TRACING** process will need to be made available to the user to view. The **TRACING RESULTS** functionality will display a list of all candidate links of high to low-level requirements based on any given filtering criteria. The user will be allowed to provide feedback on the **TRACING RESULTS** in order to improve future **TRACING**.

5.2.2 Functional Requirements

5.2.2.1 RETRO shall display the list of candidate links based on the tracing and permit feedback, filtering, and display functionality.

5.3 SAVE TRACE

5.3.1 Description and Priority

The **SAVE TRACE** function shall allow the user to save the trace result.

5.3.2 Functional Requirements

5.3.2.1 RETRO shall provide a mechanism to save the current status of trace.

5.4 COMPLETING TRACE

5.4.1 Description and Priority

The **COMPLETING TRACE** function shall allow user to finish the tracing process.

5.4.2 Functional Requirements

5.4.2.1 RETRO shall mark the project completed and not permit further changes.

6. ASSESS MODE

6.1 RUN ACCURACY & COMPLETENESS

6.1.1 Description and Priority

The user needs a mechanism that will help to determine whether the **TRACING** was successfully performed based on accuracy and completeness. An accuracy test tells the user that the candidate links provided are correctly related to the high-level requirement. Completeness test tells the user that all possible links that are listed in the candidate list are included and no low-level requirements are missing.

[1]

Functional Requirements

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

6.1.1.1 RETRO shall provide a mechanism for the user to be able to run an **ACCURACY & COMPLETENESS** check by performing a full trace on the requirements and comparing the IR method results to the given RTM. [1]

6.2 SHOW RECOMMENDATIONS LIST

6.2.1 Description and Priority

"The [user] will need to have a list of recommendations based on RETRO's **TRACING** process. **SHOW RECOMMENDATION** will display to the [user] a list of recommendation while in **ASSESS** mode. The [user] will have the capability to provide **FEEDBACK** to the RETRO system on whether or not the [user] agrees with RETRO's assessment of the linking between the low-level requirements to the high-level requirements." [1]

6.2.2 Functional Requirements

6.2.2.1 RETRO shall display/hide recommendations at the user's request.

6.3 RUN SATISFACTION ANALYSIS

Satisfaction function shows how satisfied the high-level requirement is with the given set of low-level requirements. [4]

6.3.1 Description and Priority

The Satisfaction Analysis function is checking the level of percentage for a satisfaction of the linked result with the high and low level input files. Once final output is agreed, the list of high level requirement is updated with options like "Satisfied", "Not Satisfied" and "Partially Satisfied". The status of high level requirement is color coded. Unsatisfied result can be repeated until all the high levels has been satisfied stage.

6.3.2 Functional Requirements

6.3.2.1 RETRO shall display to the user the percentage level of link. [1]

6.4 REMOVE ALL NO LINKS

6.4.1 Description and Priority

While in **ASSESS** mode, the user will need a mechanism to allow the removal of all low-level requirements that are not linked to a high-level requirement. **REMOVE ALL NO LINKS** function will provide an easy way for the user to quickly remove low-level requirements with status of "No Link".

6.4.2 Functional Requirements

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

6.4.2.1 RETRO shall remove all links marked as “No Link”

6.5 DELETE RECOMMENDED LINKS

6.5.1 Description and Priority

While in **ASSESS** mode, the user will need a mechanism that will allow the deletion of recommended links that the user feels are not correctly linked.

6.5.2 Functional Requirements

6.5.2.1 RETRO shall permit the user to select a recommended link and delete it.

6.6 COLOR-CODED SATISFACTION STATUS

6.6.1 Description and Priority

“In order to help the user to quickly identify the level of satisfaction of each candidate link, RETRO will need to provide a visual aid. **COLOR-CODED SATISFACTION STATUS** function will color code each candidate link based on its satisfaction level. The various colors will be a visual aid to the user to help quickly identify each links satisfaction level.” [1]

6.6.2 Functional Requirements

6.6.2.1 RETRO shall **COLOR-CODE THE SATISFACTION STATUS** of each given candidate link.

7. Non-Functional Requirements

7.1 Usability

U01. The RETRO System shall provide user-friendly, intuitive interface.

U02. The RETRO System shall provide an easy way to view each individual requirement.

7.2 Reliability

R01. The RETRO System shall correctly label all features such as buttons, headers, and all markers.

R02. The RETRO System shall properly spell all labels, GUI interfaces, and document for RETRO.

R03. The RETRO System shall not crash.

7.3 Performance

7.4 Supportability

7.5 Design Constraints

7.6 User Documentation Requirements

UD01. The system shall provide a help system that describes and illustrates all system functions.

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

7.7 Interfaces

7.7.1 User Interfaces

7.7.2 Hardware Interfaces

7.7.3 Software Interfaces

7.7.4 Communications Interfaces

7.8 Licensing Requirements

7.9 Legal, Copyright, and Other Notices

7.10 Applicable Standards

8. Supporting Information

8.1 Requirements/RETRO Traceability Matrix

Requirement ID	Requirement Title	RETRO coding/Function	RETRO Class
-----------------------	--------------------------	----------------------------------	--------------------

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

APPENDIX 1 – Test Cases For Assess Mode

Req ID	Requirement Description	Steps to Implement	Expected Results	Result	Bug ID
3.3.2.1	The user shall be prompted to provide the new project name.	Click on "Load RTM" prompt. Enter name of Project within "Project Name" text field.	RETRO shall prompt user to enter name for new project.	PASS	
3.3.2.2	The user shall be prompted to provide the low level document path using the standard file dialog.	Enter "Low-level document path" in text box C:\Retro-Beta\Student-Dataset\low	RETRO shall prompt user to provide file path for low-level requirements.	PASS	
3.3.2.3	The user shall be prompted to provide the high level document path using the standard file dialog.	Enter "High-level document path" in text box C:\Retro-Beta\Student-Dataset\high	RETRO shall prompt user to provide file path for high-level requirements.	PASS	
3.3.2.4	The user shall be prompted to provide the RTM file path.	Enter "RTM File Path" C:\Retro-Beta\retro\t2\result.xml	RETRO shall prompt user to provide file path for RTM.	PASS	
3.3.2.5	The user shall optionally be prompted to select an IR Method from the following list: *Vector-Space Retrieval *Keyword Extraction *Keyword Extraction-IDF	I visually verified that these three IR Methods options were made available to the user.	RETRO shall provide the user the following IR Method Options: *Vector-Space Retrieval *Keyword Extraction *Keyword Extraction-IDF	PASS	
3.3.2.6	The user shall optionally be prompted to select a Vocabulary Base of either "Both Documents" or "Low Level Document Only". The default value is "Both Documents".	I visually verified that these two Vocabulary Base options were made available to the user.	RETRO shall provide the user the following Vocabulary Base Options: *Both Documents *Low Level Document Only	PASS	

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

Req ID	Requirement Description	Steps to Implement	Expected Results	Result	Bug ID
3.3.2.7	The user shall optionally be prompted to provide Weight Option from the following list: • TF-IDF (Default) • Okapi • LTU	I visually verified that these three Weight options were made available to the user.	RETRO shall provide the user the following Weight Options: • TF-IDF (Default) • Okapi • LTU	PASS	
3.3.2.8	If the user has selected an IR Method of Keyword Extraction or Keyword Extraction-IDF, the user must also select a Weight Threshold.	First, I selected IR Method of Keyword Extraction and entered 0.25 for Weight Threshold. Then I selected IR Method Keyword Extraction-IDF and entered 0.25 for Weight Threshold. I selected one at a time both options and did not enter any values into the weight threshold & Clicked "OK".	RETRO will provide the user the ability to enter weight threshold when selecting Keyword Extraction or Keyword Extraction-IDF. If a user does not enter a weight threshold value, then a warning should prompt user to enter weight threshold value.	PASS	
3.3.2.9	The user shall optionally be prompted to provide a Thesaurus Filename.	I visually verified that "Thesaurus Option" was made available.	RETRO provide "Thesaurus Option" to user.	PASS	
3.3.2.10	The user shall optionally be prompted to provide the Stop Word file.	I visually verified that "Stop Word File" was made available.	RETRO provide "Stop Word File" option to user.	PASS	
3.3.2.11	The user shall optionally be prompted to provide a Feedback Method. "Standard Rochio" is currently the only option.	I visually verified that "Feedback Method" option was available as well check to see if the default is "Standard Rochio" available.	RETRO provide "Feedback Method" and that the default is "Standard Rochio".	PASS	

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

Req ID	Requirement Description	Steps to Implement	Expected Results	Result	Bug ID
3.3.2.12	The user shall optionally be prompted to provide the Project Mode of either "Trace" or "Assess". "Trace" is the default mode.	I visually verified that both "Trace" (default) and "Assess" modes were made available.	RETRO provide the mode options of "Trace" or "Assess". "Trace" should be the default mode.	PASS	
3.3.2.13	The User shall be able to cancel the process and return to the previous state at any time before the project is created.	I clicked on the cancelled button.	RETRO will cancel the process when prompted to cancel by the user.	PASS	
3.3.2.14	The user shall be able to create the project using the above input. The project shall be placed in "Assess" mode.	I completed all the above fields default input fields and selected "Assess" mode. Just a note there are many combinations of options when creating Assess project, so test cases should be ran to test all combinations at some point in time.	RETRO will create the Assess project when the above input default fields are completed with "Assess" mode selected.	PASS	
3.4.2.1	The user shall be able to save the current project	Go to File Menu and select Save.	RETRO shall save the current project.	FAIL	Bug 208
3.5.2.1	The user shall be able to close current project.	Go to File menu and select Close.	RETRO shall prompt the user about the current project will be saved and exit.	PASS	
3.7.2.1	The user shall be able to show XML when that function is selected.	Go to Action menu and select showXML.	A list of high and low connection is showed in XML format with a new screen.	PASS	
3.8.2.1	The user shall be able to generate report with RTM option.	Go to Action menu and select Generate Report and select checkbox for RTM.	RTM report is generated in a new screen.	PASS	
3.8.2.2	The user shall be able to generate report with RTM With Text option.	Go to Action menu and select Generate Report and select checkbox for RTM With Text.	RTM With Text report is generated in a new screen.	PASS	

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

Req ID	Requirement Description	Steps to Implement	Expected Results	Result	Bug ID
3.8.2.3	The user shall be able to generate report with Parents With No Children option.	Go to Action menu and select Generate Report and select checkbox for Parents With No Children.	Parents With No Children is generated in a new screen.	PASS	
3.8.2.4	The user shall be able to generate report with Parents With No Children With Text option.	Go to Action menu and select Generate Report and select checkbox for Parents With No Children With Text.	Parents With No Children With Text is generated in a new screen.	PASS	
3.8.2.5	The user shall be able to generate report with Children With No Parents option.	Go to Action menu and select Generate Report and select checkbox for Children With No Parents.	Children With No Parents is generated in a new screen.	PASS	
3.8.2.6	The user shall be able to generate report with Children With No Parents With Text option.	Go to Action menu and select Generate Report and select checkbox for Children With No Parents With Text.	Children With No Parents With Text is generated in a new screen.	PASS	
3.8.2.7	The user shall be able to generate report with Not Satisfied Requirement option.	Go to Action menu and select Generate Report and select checkbox for Not Satisfied Requirement.	Not Satisfied Requirement is generated in a new screen.	PASS	
3.8.2.8	The user shall be able to generate report with Not Satisfied Requirement With Text option.	Go to Action menu and select Generate Report and select checkbox for Not Satisfied Requirement With Text.	Not Satisfied Requirement With Text is generated in a new screen.	PASS	
3.9.2.1	The user shall be able to designate whether the filter process applies to the selected high level document element (LOCAL) or to all high level document elements (GLOBAL).	On Trace or Assess Tab, click on GLOBAL or LOCAL.	Option clicked on is selected.	PASS	

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

Req ID	Requirement Description	Steps to Implement	Expected Results	Result	Bug ID
3.9.2.2	The user shall be able to filter the list of low-level requirements by setting a cutoff point for the weight between 0 and 1.0.	In filter textbox, type a number between 0 and 1. Select filter slider bar and move the slider position.	Number is properly formatted, and slider bar is placed at appropriate position.	PASS	
3.9.2.3	The user shall be able to designate that the top N elements to be displayed.	Select the "Show Top" radio option, and enter a positive number in the textbox.	Unable to Test	N/A	
3.9.2.4	The user shall be able to activate a Kill Hidden Links function to remove all elements not currently displayed.	Click on the "Kill Hidden Links" button.	Unable to Test	N/A	
3.10.2.1	The user shall be able to sort the low level documents by Requirement	Click on the "By Requirement" radio option	Unable to Test	N/A	
3.10.2.2	The user shall be able to sort the low level documents by Weights	Click on the "By Weights" radio option	Unable to Test	N/A	
3.10.2.3	The user shall be able to sort the low level documents by Document Order	Click on the "In Document Order" radio option	Unable to Test	N/A	
3.11.2.1	The user shall select a low level requirement and identify the link status from the following list: - Link - No Link - Default	Right click on the selected low level requirement. Select Change -> Link, then repeat with Change -> No Link and Change -> Default	Unable to Test	N/A	
6.1.2.1	The user shall be able to run an AccuracyCompleteness to perform a full trace on the requirements and comparing RETRO IR method results to the given result from the given RTM.	Go to "Assess" tab and click AccuracyCompleteness button.	Recommended links will be present if there are any candidate links in all the result set.	FAIL	Bug 251

RETRO - release V.Beta	Version: 1.0
Functional Specification	Date: 09/11/06
SVVRL-RETRO-func-spec-v1.doc	

Req ID	Requirement Description	Steps to Implement	Expected Results	Result	Bug ID
6.2.2.1	RETRO shall display/hide recommendations at the user's request.	Click on the "Show Recommendations" / "Hide Recommendations" button.	Recommendations GUI is displayed.	PASS	
6.3.2.1	The user shall be able to obtain the percentage level of satisfaction of the linking of the given low-level requirements to the high-level requirement.	Select Load RTM and insert appropriate input. Run Satisfaction button and view all the various percentage levels for each link.	Display to the user the percentage level of satisfaction of the linking of the given low-level requirements to the high-level requirement.	Fail	Bug 208
6.4.2.1	The user shall be able to Remove All No Links.	Go to Action and select Remove All No Links.	RETRO shall remove all links marked as "No Link"	PASS	
6.5.2.1	RETRO shall permit the user to select a recommended link and delete it.	Select a low level requirement, right click and select "Delete".	Unable to Test	N/A	
6.6.2.1	The user shall be able to obtain color-code satisfaction status of each given low-level requirement link with the high-level requirement.	Go to Assess tab and look for color-coded links.	Color-coded links will be available if there are any links between the low-level requirement and the high-level requirement.	PASS	